



Procedures for Laser Cutting

for

Mechanical Project 478

Mechatronic Project 478

Postgraduate Students

The use of laser cutting as a manufacturing process is encouraged and can enhance your design if cleverly used. The Department of Mechanical & Mechatronic Engineering often makes use of Fabrinox for the manufacture of laser cutting parts, bending and welded assemblies. This document provides design tips and the steps to follow to creating the required drawings and files. The example is that of a welded assembly. If you only require the manufacturing of a single part, then the same procedure is followed as described below for one of the parts.

The first step is to design the parts. The example used, is the design of a bracket to fix a rooftop tent to the back of a bakkie, Figure 1. The welded assembly consist of two parts, Bracket Part 1 and Bracket Part 2. The welding assembly is shown in Appendix A, drawing BA01, with the details of Bracket Part 1 and Bracket Part 2 on drawings BA02 and BA03 respectively. Note that drawings BA02 and BA03 consist of two pages each (1 van 2) and (2 van 2).

1. Design Tips

- Make use of Autodesk Inventor's sheet modelling.
- The cost of the final part is determined by the material cost and the manufacturing cost (laser cutting, bending and welding). The material cost is determined by the material type (stainless steel, mild steel, aluminium) and the plate thickness. The cost of laser cutting is determined by the type of material, material thickness, accumulated length of the cut and the number of piercings (the number of times a new cut is started, for example a hole).
- See the Fabrinox website for available materials, www.fabrinox.com.
- See the Fabrinox bending details, Appendix B, for minimum bend radii and allowable dimensions.
- The diameter of a hole can not be smaller than the material thickness. If such a small hole is required, it has to be machined afterwards. In such a case Fabrinox can however, engrave the centre position of the whole on the material. Therefore holes too small for

laser cutting should not be included in the drawing and “dxf” model or a note should be added to the drawing indicating which holes should only be engaged.

- If the parts are assembled using nuts and bolts, make the holes slightly larger than the bolt size. For most sizes a 0.5 mm to 1 mm tolerance on the diameter should be more than sufficient.
- Also, when parts are bolted together make use of slots rather than holes where applicable. Slots allow for adjustments to be made and can compensate for manufacturing tolerances. However, long slots can decrease the torsional stiffness of a component and it is recommended to rather use multiple slots with the length of each not more than 5 times the width.
- It is easier to add holes to the laser cut component than to drill holes afterwards. If functionally possible, add holes to positions where other components might be bolted onto in future.
- Do not add holes too close to a bend radius. The material will warp and the bend will not be accurate in the close vicinity of the hole.
- If the diameter of a hole needs to be accurate (for example sliding and press fits) it needs to be machined afterwards. Therefore, the diameter of the hole on the laser cut drawing should be 1 mm to 2 mm smaller than that of the final design which will allow for machining in our workshop.
- The same applies for a tapped hole. The hole on the laser cut drawing should be smaller than the bolt diameter to allow for machining afterwards in our workshop. Take the thread depth into account. For example, the outside (major) diameter of a M10 bolt with a 1.5 mm pitch is 10 mm and the root (minor) diameter is 8.1 mm. In this case the laser cut diameter should be less than 8.1 mm, typically 7 mm. This will allow the workshop to first drill the hole to the desired diameter and then tap the hole.
- If weight is important, add holes to the part if it warrants the additional cost involved. It also creates a more aesthetically pleasing part.
- Add fillets to all the sharp edges. Small fillets, typically 1 mm or 2 mm radius, might have no effect on the functionality of the part, but the part is safer and looks better at no additional cost (and less piercings).
- By bending the material, the overall stiffness of the component can be increased. Clever use of bends can result in a light weight component with high bending stiffness. See the Fabrinox bending details, Appendix B, for minimum lip lengths.
- Make use of clever designs such as those shown in Figure 2. Using slots can help to locate one component relative to another. This also applies to welded assemblies. When designing “boxes”, try to keep the number of individual side panels as low as possible, rather bend the plate to form the sides.
- See Mr Cobus Zietsman or Mr Ferdi Zietsman before you finalise your drawings to discuss the design and manufacturing process.



Figure 1 – Final bracket design

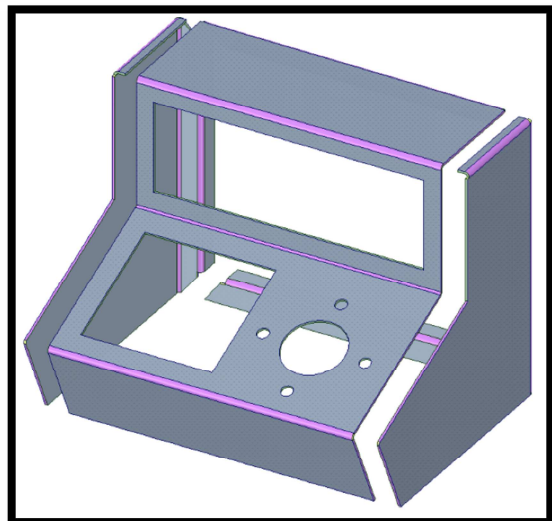
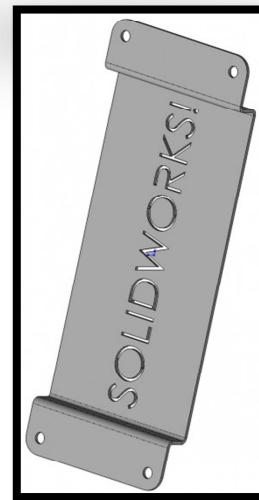
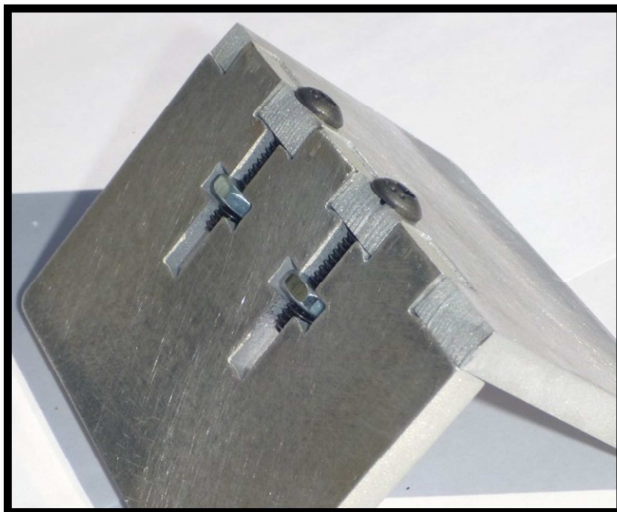


Figure 2 – Clever designs

2. Drawings

Once you have completed the design, the correct drawings and files need to be sent to Fabrinox for a quote. The drawing standards as prescribed by the Department of Mechanical and Mechatronic Engineering apply.

2.1 Assembly Drawing

If your design includes more than one component that needs to be assembled (welded), you need to include an assembly drawing. The workshop (Mr Zietsman) will indicate if the assembly and/or welding will be performed by Fabrinox or if it will be done at the M&M Workshop (MMW). If assembly and welding will be done at MMW you do not need to send an assembly drawing to Fabrinox, only the detail design of each part.

An example of an assembly drawing is shown in Appendix A, drawing BA01 and should include the following:

- Enough views (front, top, side, etc.) to show all the detail necessary for assembly/welding.
- An isometric view is not required, but helps with visualising the part, especially if it has a complicated shape.
- If it is a welded assembly, indicate where welds should be made. You do not have to provide weld specifications (fillet height, etc.), the welder should know what kind of weld is required for the specific material and plate thickness.
- Parts list with reference to the individual parts, including the required quantity of each and a drawing reference.

2.2 Part Detail Drawing

This drawing provides the details of each final part. Drawings BA02(1 of 2) and BA03(1 of 2) are examples and should include the following:

- Enough views (front, top, side, etc.) to show all the detail necessary for assembly/welding.
- An isometric view is not required, but helps to visualising the part, especially if it has a complicated shape.
- Only major and important dimensions should be provided. The drawing pack should be accompanied by a “dxf” file. This file will contain all the dimensions of the unfolded (2D) part as well as the bending lines and therefore not all dimensions are needed here. However, the “dxf” file does not contain the dimensions of the part once folded, and therefore the dimensions that are important in terms of function should be provided. Also, the manufacturer needs to make sure that the component was accurately bent and for this you need to provide overall dimensions, for example the 88 mm height and 84 mm width of the part in drawing BA02(1 of 2). In this case, a load bar fits into the 34 mm wide horizontal slot, so this is an important dimension and therefore specified. Also, Bracket Part 2 needs to be welded to Bracket Part 1, and therefore the 84 mm width is also specified on drawing BA03(1 of 2). If a part

needs welding, it should be indicated as on drawing BA03(1 of 2). The smaller holes can be dimensioned so that the manufacturer can see if it can be laser cut or not.

- Make sure to specify the material type and plate thickness.
- If this component is part of an assembly, the quantity is not specified here, only on the assembly. If the component is not part of an assembly, the quantity should be specified on the detail drawing.

2.3 Bending Schedule

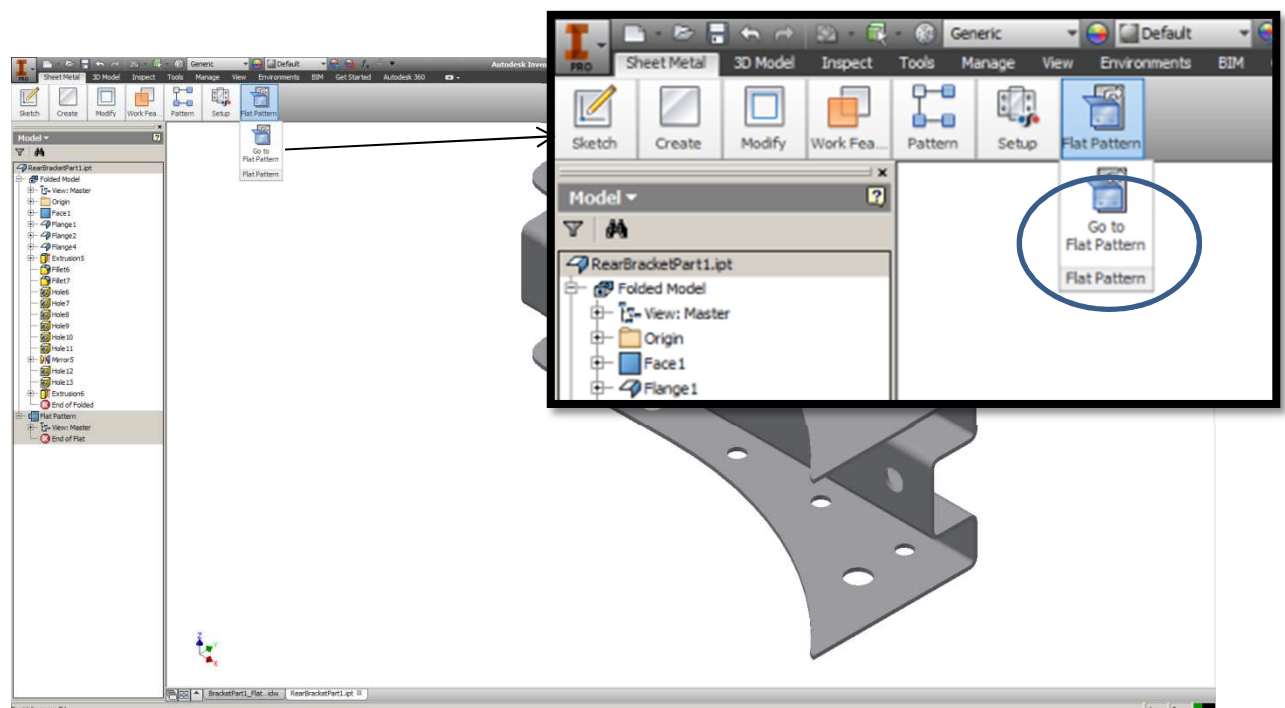
The “dxf” file provides the outline of the unfolded part and the bending lines in electronic format but not the direction and angle of the bends. Therefore a bending schedule needs to be prepared. Examples are shown in drawings BA02(2 of 2) and BA03(2 of 2) and should include:

- The unfolded part including the outline and bending lines.
- Provide the overall dimensions (size of material that will be needed if the part were to be boxed in). This helps the manufacturer to easily get a feeling of the size of the part and can be used to make sure that the “dxf” file was produced using the correct scale.
- Although the bending lines are included in the “dxf” file, it should also be added to this drawing so that the operator can see where to bend.
- For each bend, indicate the direction and angle of the bend.

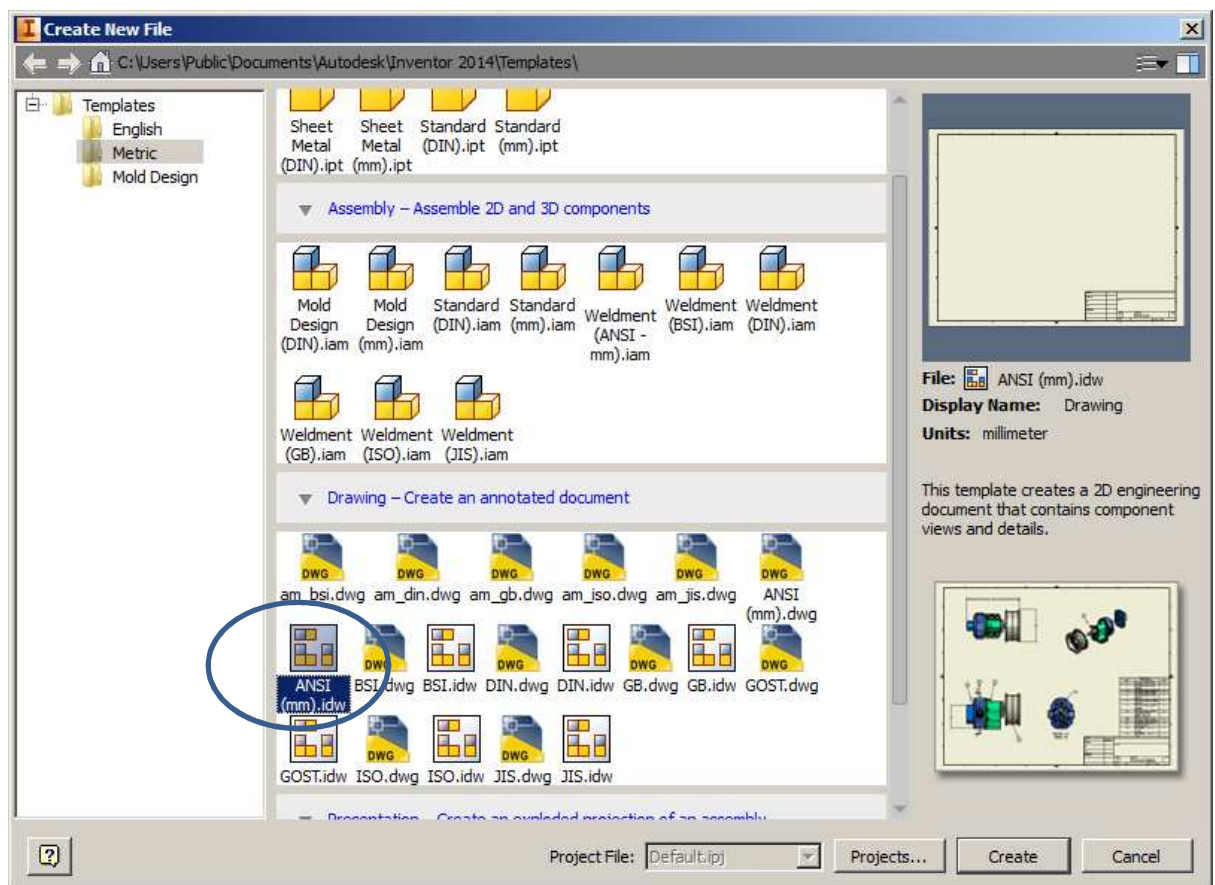
3. DXF File

A “dxf” file of each unfolded (flat pattern) part should accompany the drawing pack. This file is directly used by the manufacturer to laser cut the part. It is important to create the “dxf” file with the unfolded part at a scale of 1:1 otherwise the manufactured part will be scaled. The following steps should be used (Autodesk Inventor Professional 2014 was used, the procedure might be different in other versions, but the steps still apply):

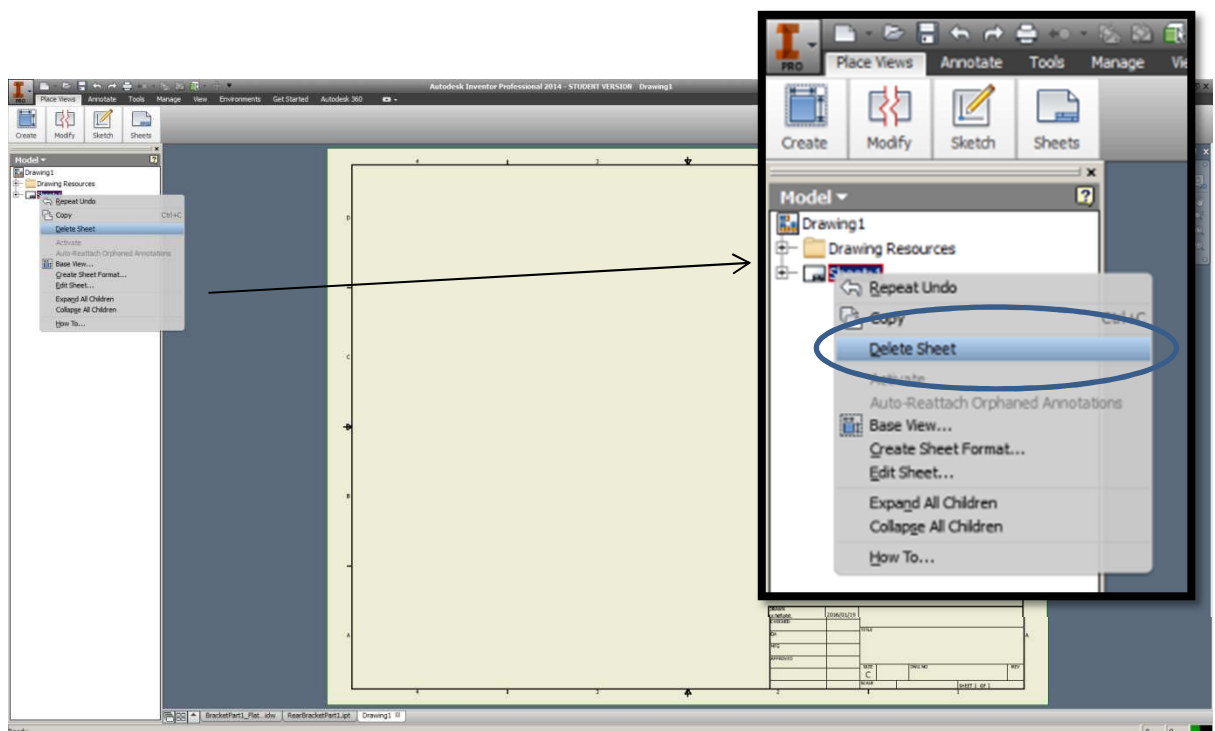
1. Create a flat pattern of the part by clicking on “Flat Pattern”



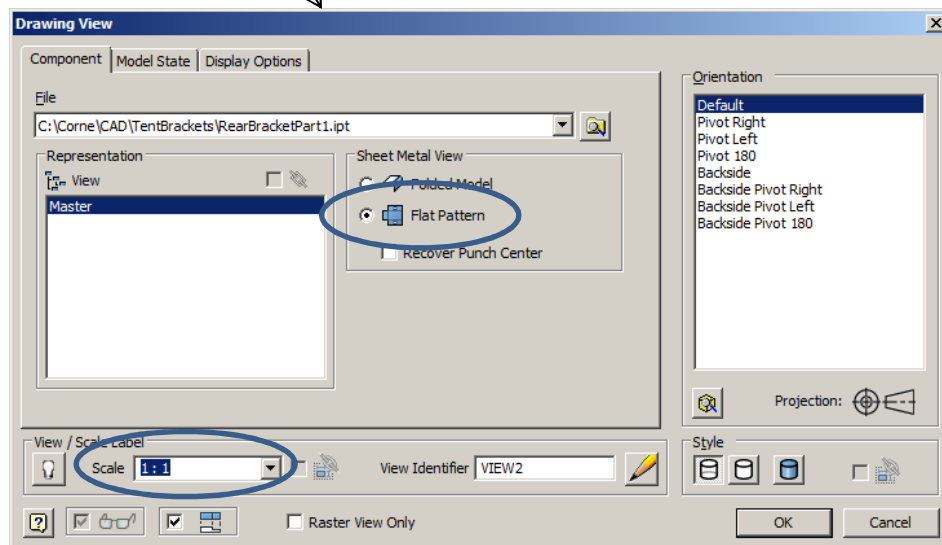
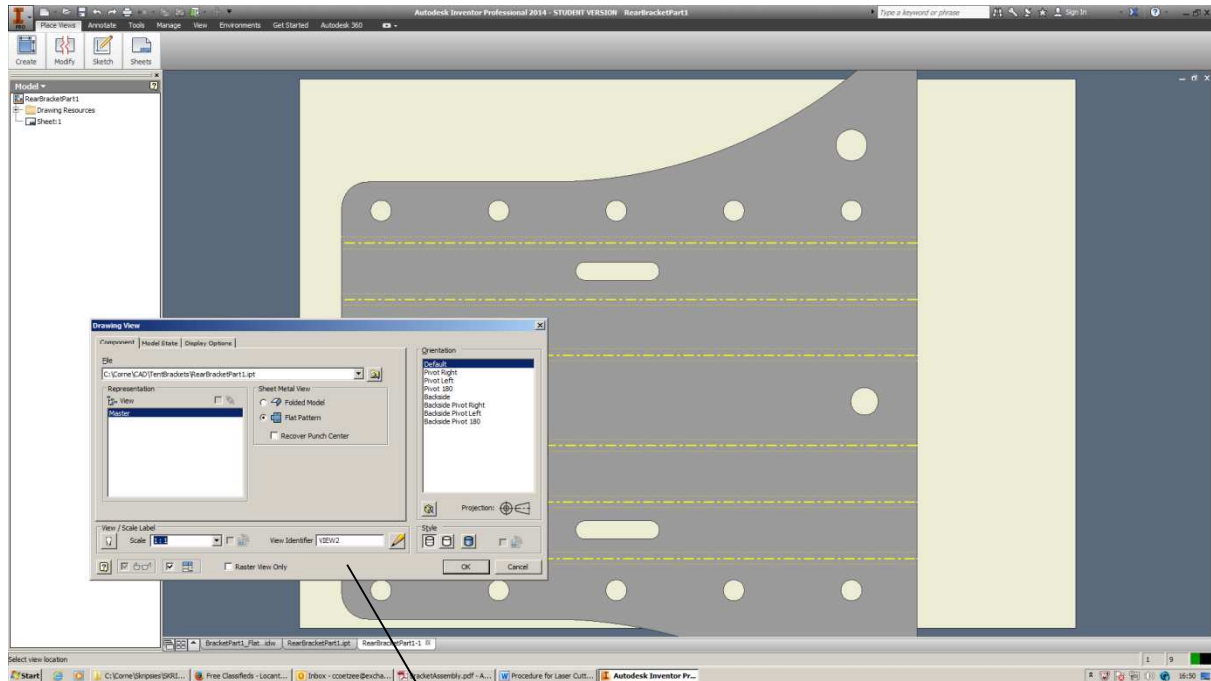
2. Create a new ANSI(mm).idw file



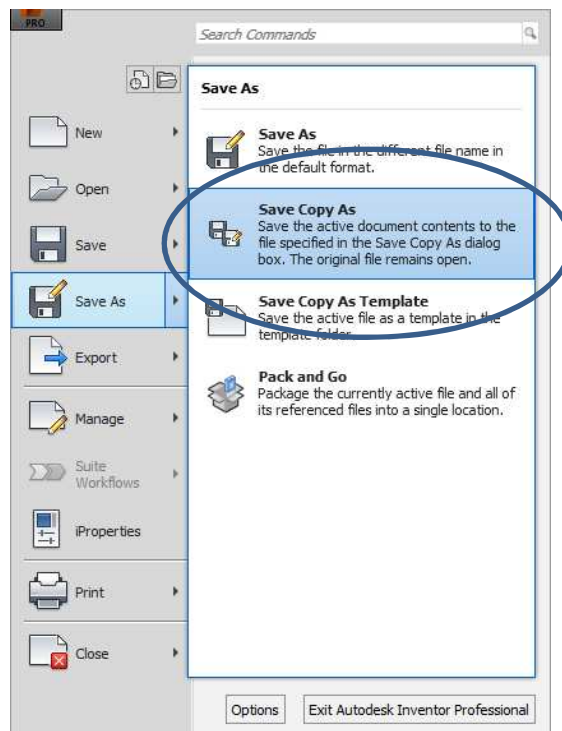
3. This will create a new sheet including a title block. In the left hand column, delete the sheet. This removes the title block.



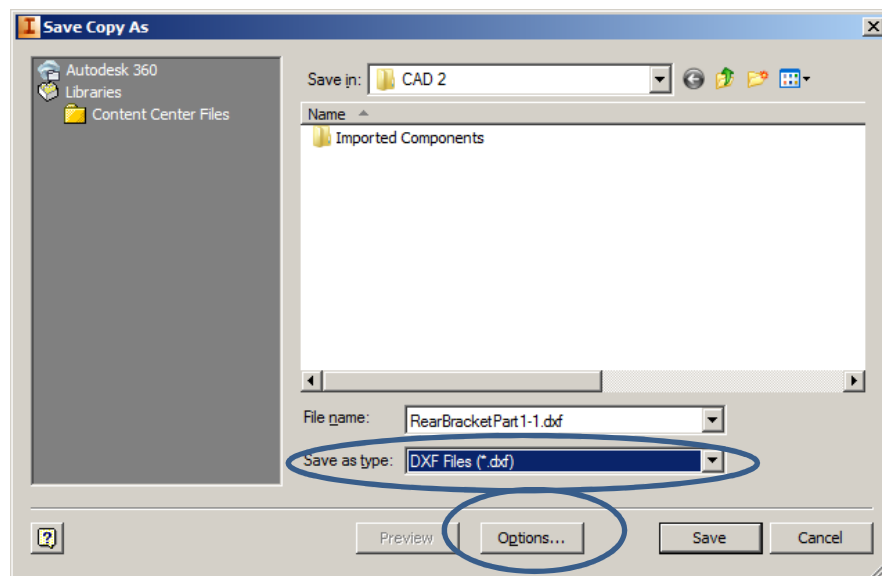
4. Select “Create” -> “Base” -> “Flat Pattern” and scale 1:1 to create a new view of the unfolded part. The scale of the part relative to the page does not matter. Even if the part extends over the edges of the page.



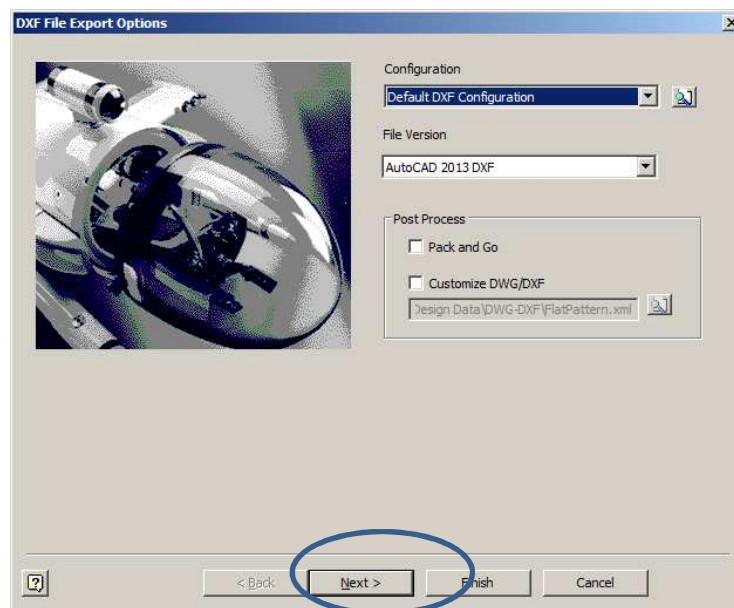
5. Select “File” -> “Save As” -> “Save Copy As”



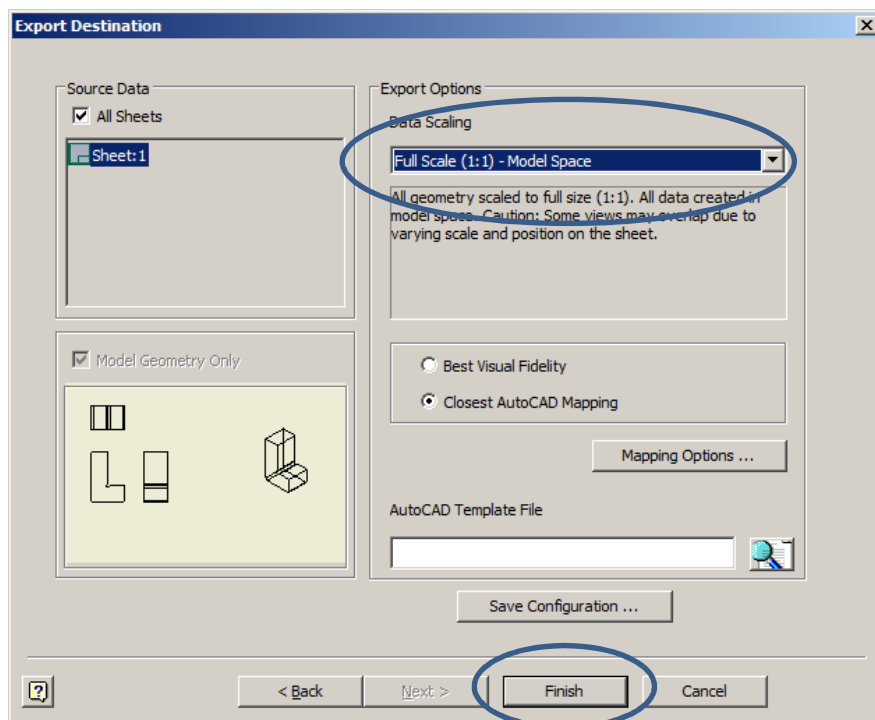
6. Under “Save as type” select “DXF Files (*.dxf)” and click on “Options”



7. Keep the default setting and click on “Next”



8. Under “Data Scaling” select “Full Scale (1:1) – Model Space” and click “Finish”. Select a folder and appropriate file name to save the “dxf” file. This file needs to be send with the drawing pack for a quote.



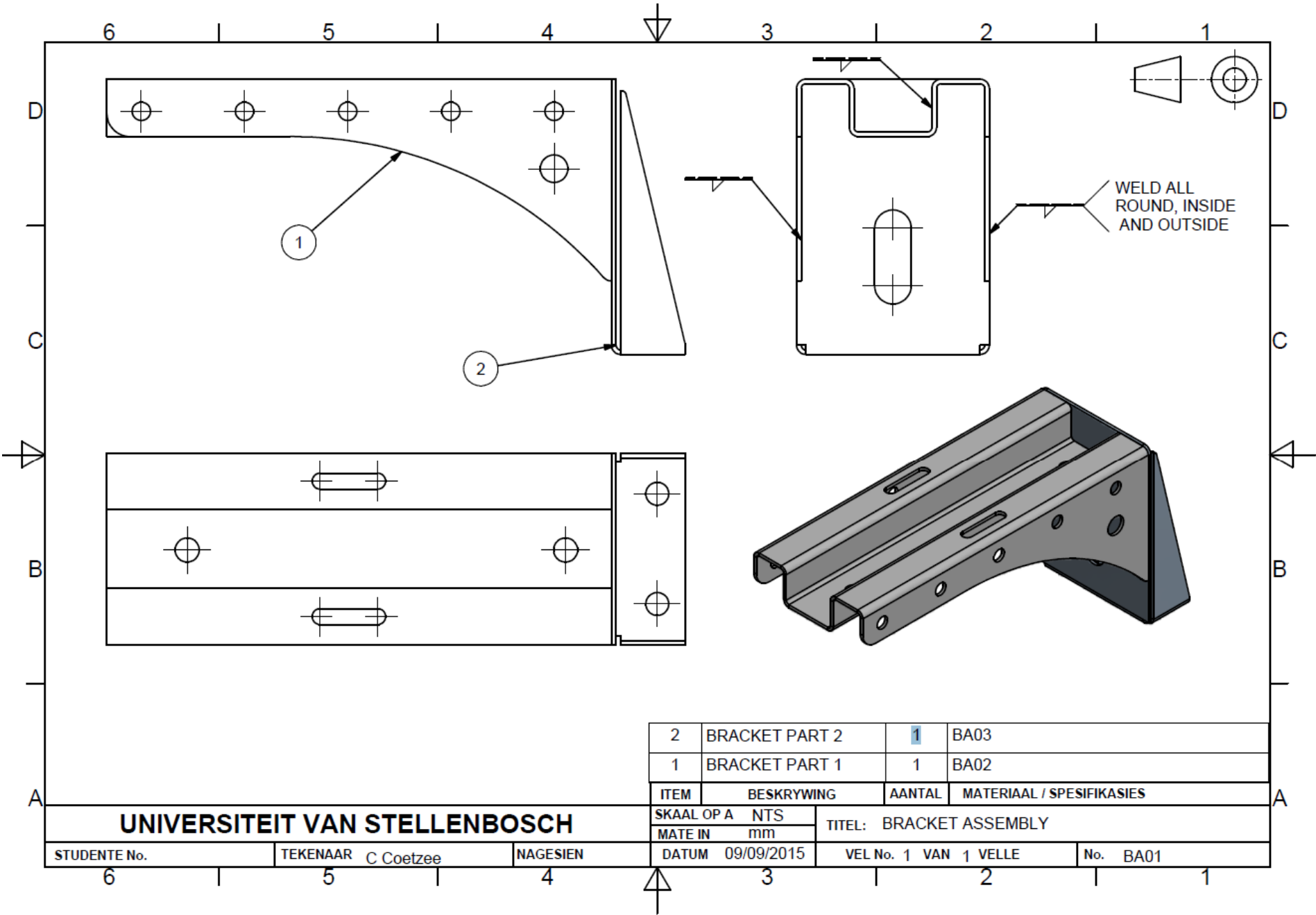
9. This file can again be opened to make sure it is correct and to the correct scale. When opening the file, make sure to select the correct “Files of Type” otherwise the “dxf” files will not show. Use the measuring tool to measure a known dimension to check the scale.

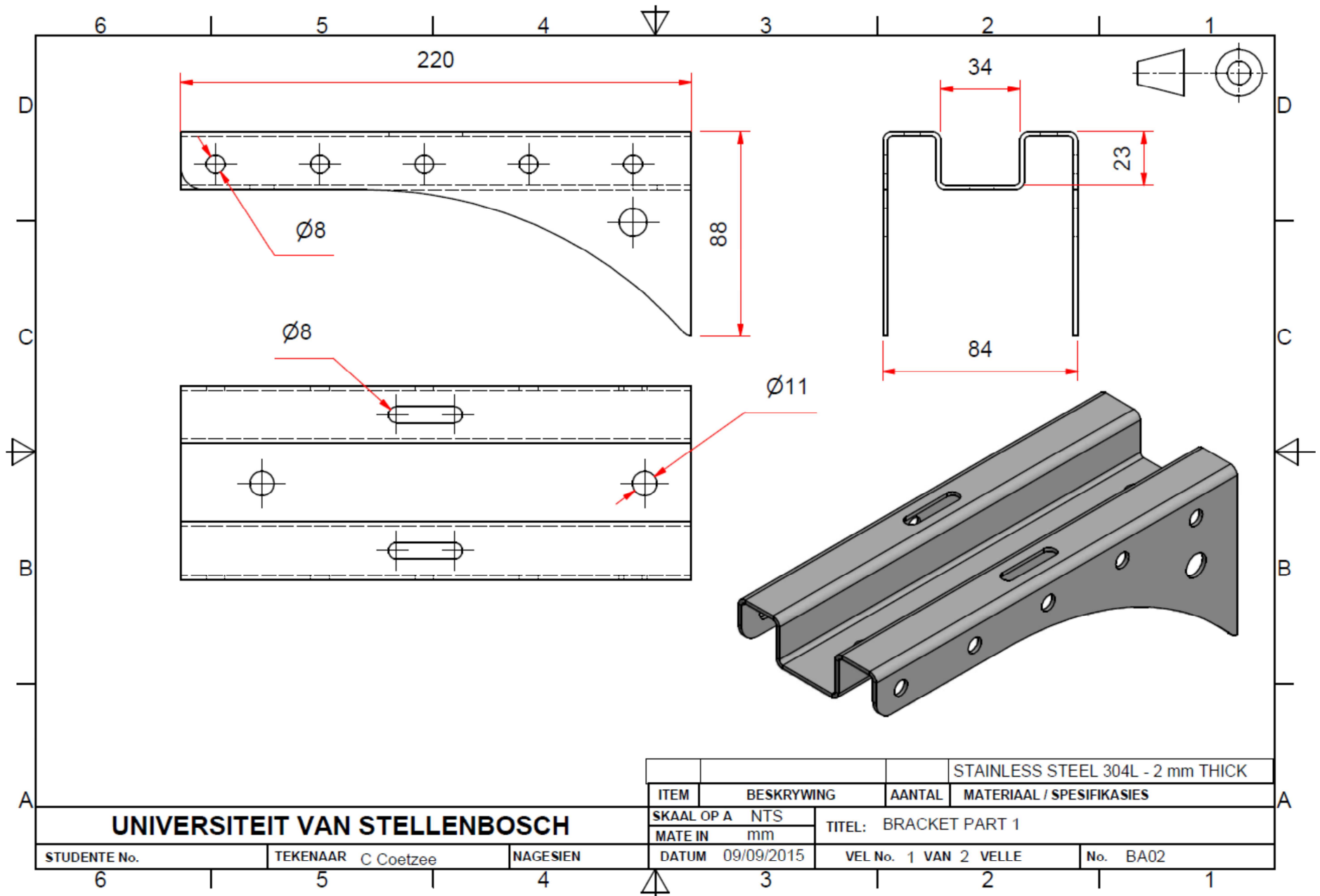


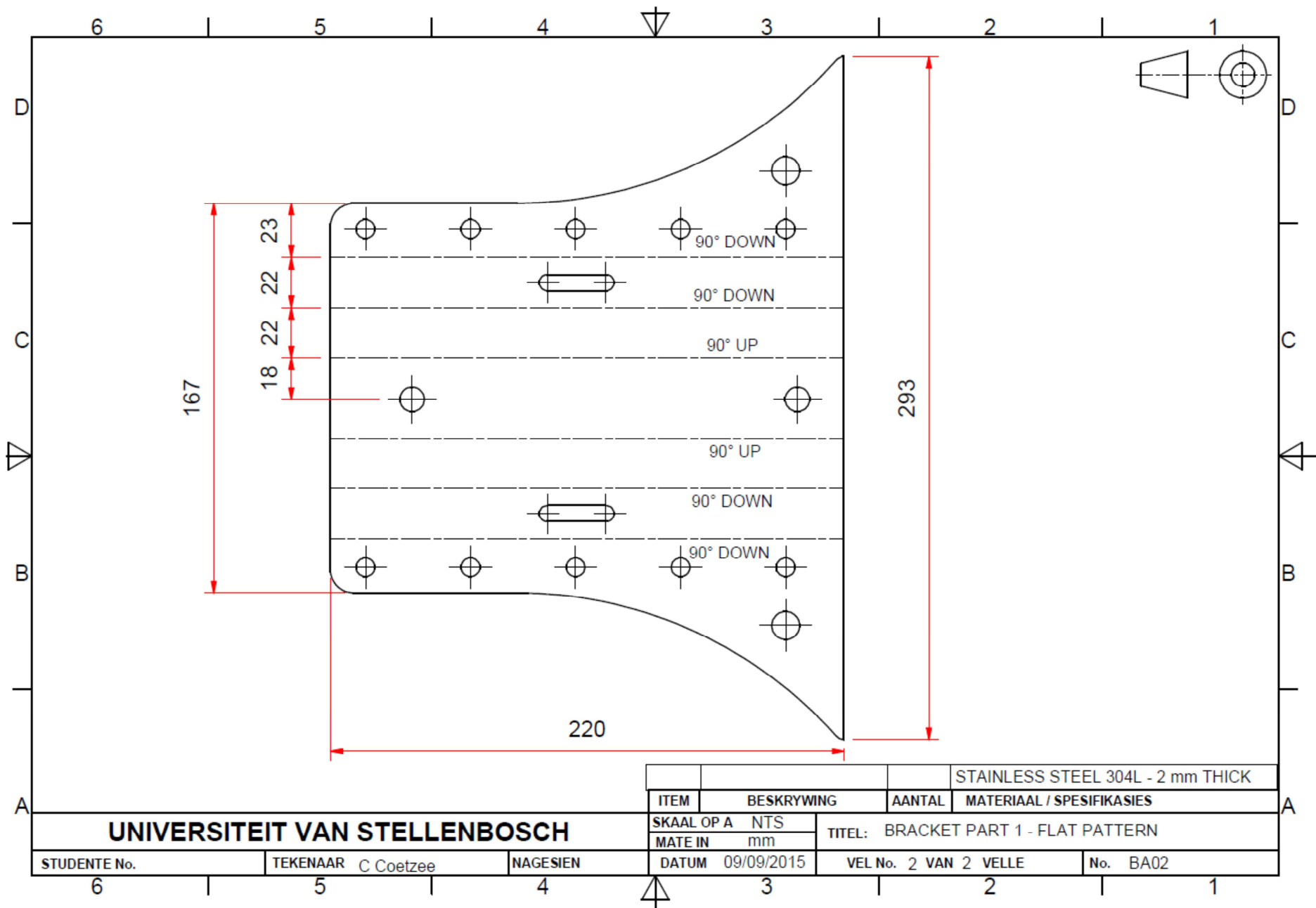
4. Design Process and Getting a Quotation

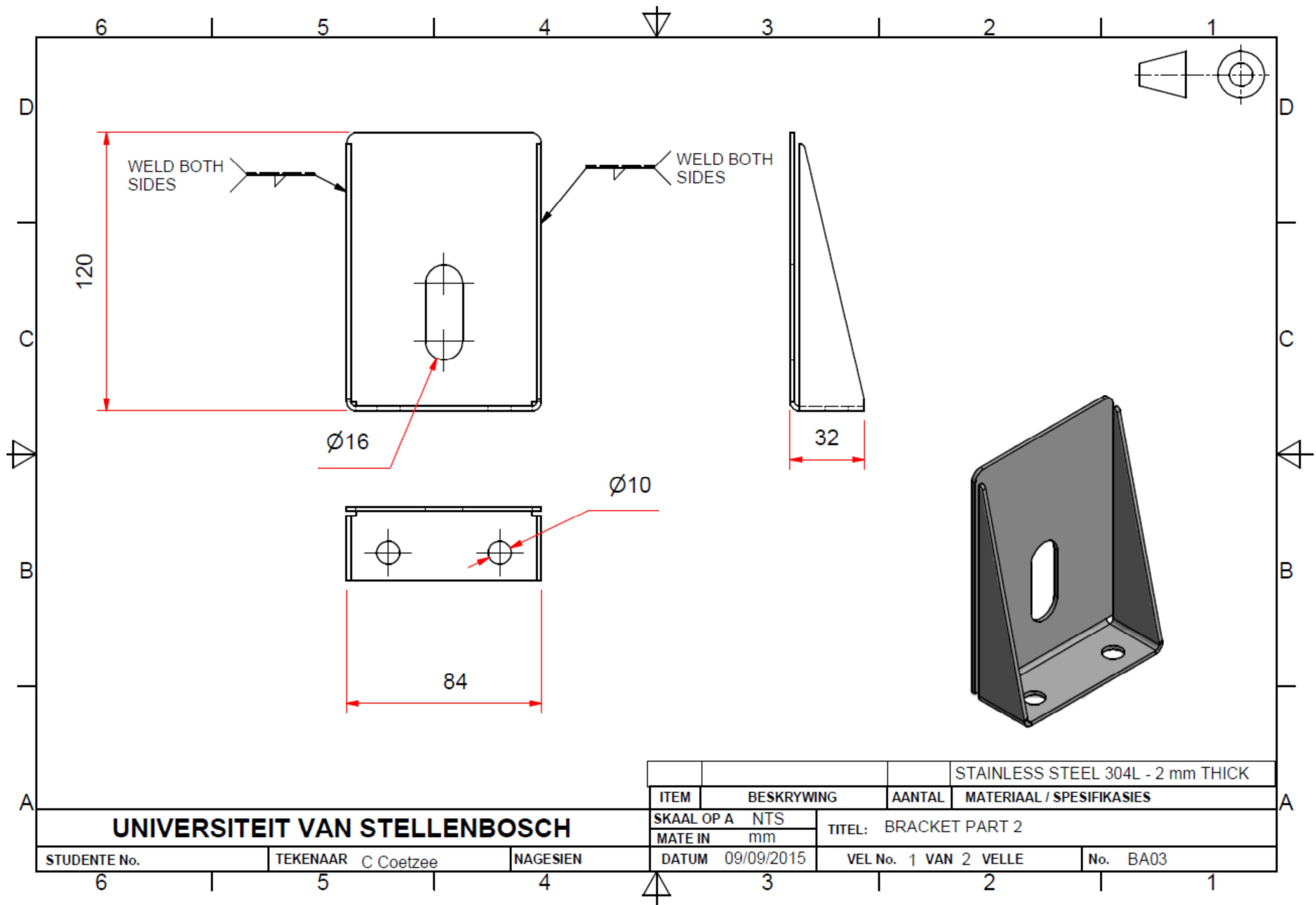
1. Early on during the final design process, take your drawings to Mr Zietsman to discuss the design and manufacturing process. He will indicate which components should be laser cut and which will be manufactured in-house.
2. Prepare the drawing pack in “pdf” format and the “dxf” file(s) as described above.
3. Take these files on a USB drive to Mr Nathi Hlwempu or email it to him at hlwempu@sun.ac.za.
4. Mr Hlwempu will look at your drawings and “dxf” files to make sure they are correct and suggest changes where necessary.
5. Once everything is correct, Mr Hlwempu will email the drawing pack and the “dxf” files to Fabrinox for a quotation.
6. When Mr Hlwempu has received the quotation from Fabrinox, he will email it to you.
7. Take the quotation to your study leader for approval. He/she should sign on the quote.
8. Take the signed quote to Ms Merentia Meyer (Room 2107) to place the order.
9. The items will be delivered to the workshop. Keep in contact with Mr Hlwempu to see when your order has arrived. Relatively simple designs such as only laser cutting of small components and a small number of bends can take between 1 and 2 weeks to be delivered. More complex designs, large items and welded assemblies might take longer.

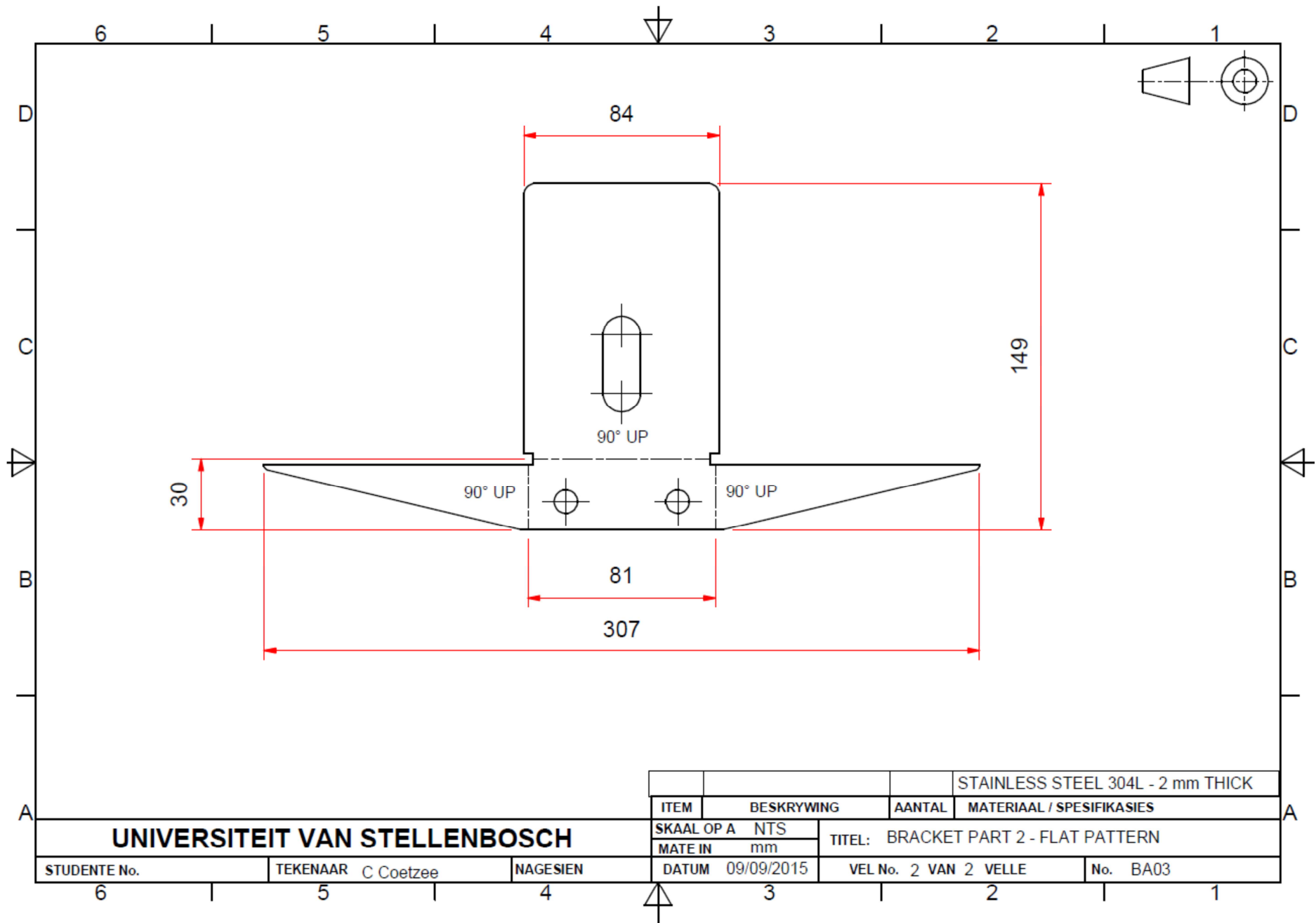
APPENDIX A – Example: Welded Bracket Assembly











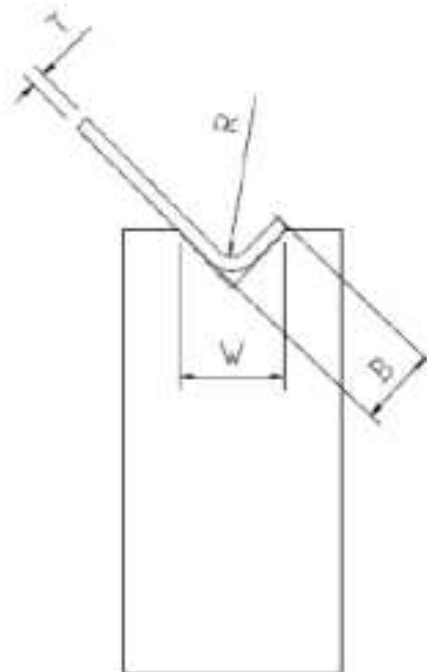
APPENDIX B – Fabrinox: Bending Details

(also available at www.fabrinox.com)

Document No. 824W008

Revision No. 02

Bending Details



V-BLOCK	BLOCK WIDTH	PLATE	MIN BEND	STAINLESS STEEL		MILD-STEEL		ALUMINIUM	
				1/RADIUS R	NEUTRAL FACTOR	1/RADIUS R	NEUTRAL FACTOR	1/RADIUS R	NEUTRAL FACTOR
6	20	0.9	4.5	1	0.4	1	0.4	1	0.4
8	20	1.2	6	1.3	0.4	1.3	0.4	1	0.4
12	25	1.6	9	1.9	0.4	1.9	0.4	1	0.4
16	30	2 & 2.5	12	2.6	0.4	2.6	0.4	1	0.4
24	35	3	18	4.3	0.3	4.3	0.4	1	0.4
40	35	4 & 4.5 & 5	30	7.1	0.29	7.1	0.4	1	0.4
50	75	6	37.5	8.8	0.33	8.8	0.4	4	0.4
70	85	8 & 10	52.5	11	0.4	11	0.4	4	0.4
100 (HELI)	120	12	75	16	0.4	16	0.4	4	0.4

Parts that are rolled must get a neutral factor of 0.505

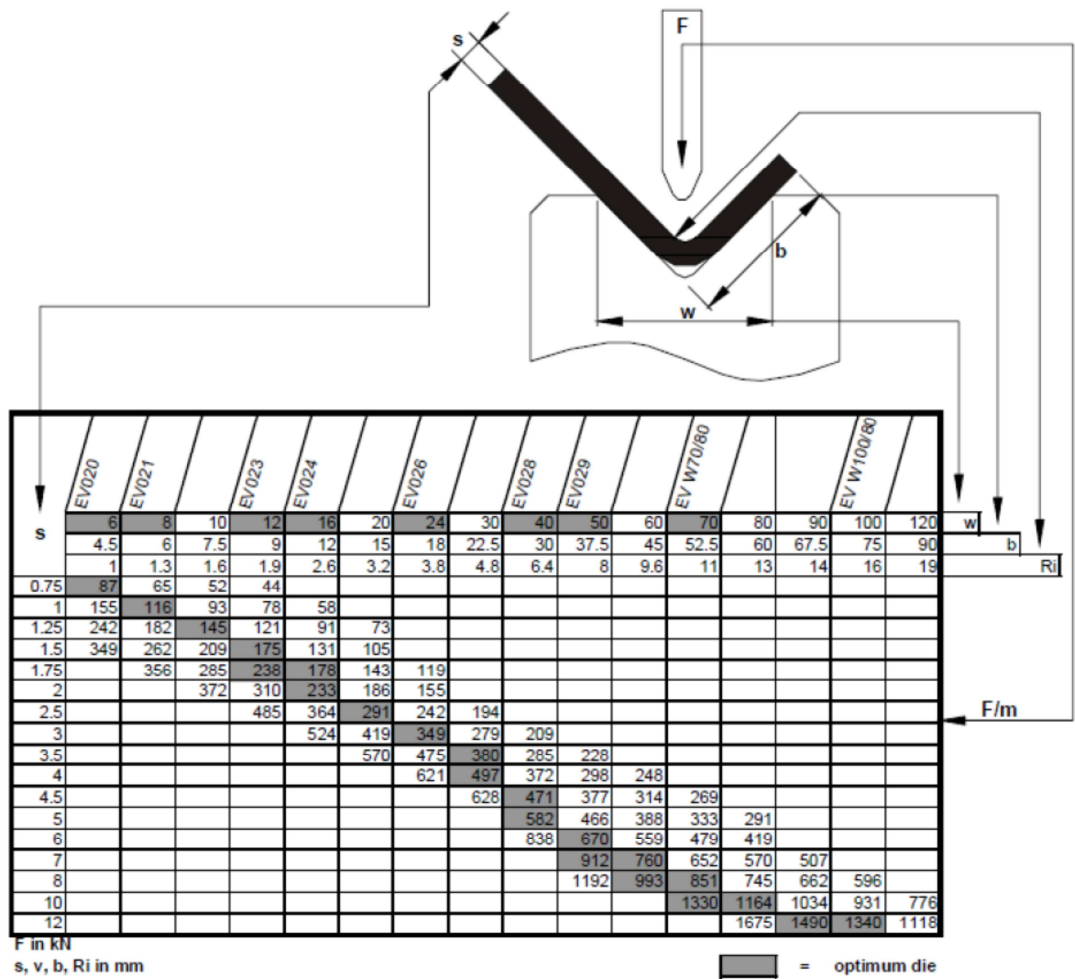
Press Force Table

824W008.02

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UNCONTROLLED if printed and not "CONTROLLED COPY" stamped in RED.
Not suitable for use if UNCONTROLLED.

Necessary press force for 90° air bending.
Material 700 N/mm², Stainless Steel.



Necessary press force for 90° air bending.
Material 400 N/mm², Mild Steel.

